Calibrated Uncertainty Quantification for QKD Anomaly Detection, with a Scale-Limited Study of Amplitude-Encoding Kernels

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***Abstract*—Quantum Key Distribution (QKD) enables information-theoretically secure communication but remains vulnerable to implementation-level attacks that exploit hardware imperfections. We present HYQAD [1], a calibrated uncertainty quantification framework for QKD anomaly detection that combines a Bayesian Gaussian Process classifier with quantum-inspired feature engineering. On synthetic QKD telemetry spanning four attack scenarios (Photon-Number-Splitting, Detector Blinding, Trojan Horse, Intercept-Resend), our deployed inference path—a Bayesian GP alone—achieves 0.994 ± 0.002 accuracy with sub-10ms latency and calibrated predictive uncertainty enabling selective prediction: at a modest uncertainty threshold ($u < 0.3$), accuracy reaches 1.000 on 89.7% of the test set with the remainder deferred for human review. We include a rigorous, scale-controlled study of a 3-qubit quantum kernel SVM via PennyLane [19]. A matched-n ablation—training all models at identical sample sizes—reveals that the quantum kernel provides no benefit at any tested training scale on QKD data, due to $\mathcal{O}(n^2)$ circuit simulation cost limiting training to $n \leq 25$ samples on classical hardware. We identify three conditions under which quantum kernels could become competitive (large n, classically-hard feature separations, hardware-accelerated circuits) and provide a controlled experiment demonstrating quantum kernel advantage on a classically-hard synthetic problem, substantiating this forward-looking claim. Sensitivity analysis across attack perturbation magnitudes and an adversarial robustness evaluation further characterize detector behavior under realistic operating conditions. Random Forest achieves perfect separation (1.000 on all metrics), indicating the synthetic data lacks real-world noise complexity; experimental validation on physical QKD testbeds is essential. Our primary contribution is a principled uncertainty quantification pipeline for QKD security monitoring, with the quantum component documented as a scale-limited component under active investigation.**

***Index Terms*—Quantum Key Distribution, Anomaly Detection, Gaussian Processes, Uncertainty Quantification, Quantum Machine Learning, Physical-Layer Security**

[1]The name is retained for continuity from our prior architecture in Fig. 1; the system's primary contribution is calibrated uncertainty quantification, not hybrid quantum–classical inference.

I. Introduction

Quantum Key Distribution (QKD) leverages the no-cloning theorem and measurement-induced disturbance to establish shared cryptographic keys with information-theoretic security [1], [2]. While protocols such as BB84 [1] and decoy-state variants [3] are theoretically secure, practical implementations introduce side channels that adversaries can exploit [4], [5].

Implementation attacks on QKD systems include Photon-Number-Splitting (PNS) [6], Detector Blinding [7], [8], Trojan Horse [9], and Intercept-Resend [10] strategies. These attacks manifest as anomalies in channel telemetry—QBER, photon count rates, detection efficiency, and background noise [11]. Classical machine learning approaches for detecting such attacks include Support Vector Machines [12], Random Forests, and neural networks [13]. However, these methods provide point estimates without uncertainty quantification—a critical limitation for security applications where false negatives carry high cost and operators need calibrated confidence for decision-making.

This paper makes three contributions:

1) **Calibrated uncertainty quantification for QKD anomaly detection.** We present HYQAD, a framework whose deployed inference path (a Bayesian Gaussian Process classifier) achieves 0.994 ± 0.002 accuracy on synthetic QKD telemetry, with calibrated predictive uncertainty enabling selective prediction that trades coverage for accuracy. This is the paper's primary, operationally deployable contribution.
2) **A rigorous, honest evaluation of quantum kernels at current scales.** We provide a matched-n ablation—training all models at identical sample sizes—that isolates the quantum kernel's contribution from confounding training-set-size effects. We show that a 3-qubit quantum kernel SVM, limited to $n \leq 25$ training samples by $\mathcal{O}(n^2)$ circuit simulation cost on classical hardware, provides no benefit at any tested training scale on QKD data. We further demonstrate quantum kernel

advantage on a classically-hard synthetic problem, establishing that the approach is sound in principle but currently scale-limited.

3) **Comprehensive characterization under realistic conditions.** We perform sensitivity analysis across attack perturbation magnitudes, adversarial robustness evaluation against adaptive evasion, and per-attack-type analysis with uncertainty-aware deferral, providing a thorough baseline for future QKD security monitoring research.

II. Related Work

Detection of QKD implementation attacks has been approached through hardware-level monitoring [20], [21], statistical process control [22], and machine learning [14]. Ren et al. [12] applied random forests and SVM to QKD protocol selection, while Liu and Zhang [13] used deep belief networks for multi-attack classification. These methods provide point estimates without calibrated uncertainty—a gap we address with Bayesian inference. To our knowledge, no prior work has applied uncertainty-aware anomaly detection specifically to QKD security monitoring; our work is the first to provide calibrated predictive probabilities that directly inform operator decision-making.

Quantum kernel methods encode classical data into quantum Hilbert spaces via parametrized circuits [15]. The resulting kernels can realize feature maps that are classically intractable [17], [23]. Recent work has characterized the computational boundaries of quantum kernel advantage [23], showing that the benefit depends critically on the alignment between the quantum feature map and the data's structure. Our work adapts these insights to QKD anomaly detection using amplitude encoding as a fixed quantum feature map—a "quantum-enhanced" rather than fully quantum approach—and provides an honest assessment of when (and when not) the quantum kernel adds value.

Bayesian anomaly detection [24] offers principled uncertainty quantification. Gaussian Processes [25] provide posterior distributions over latent functions, naturally producing predictive variances that indicate out-of-distribution samples. Our framework operationalizes this uncertainty as a confidence metric for security operators, and we validate its calibration properties on QKD data.

III. Methodology

Fig. 1 illustrates the HYQAD architecture: (i) physics-driven QKD channel simulation with four attack models; (ii) quantum-inspired feature engineering (amplitude encoding, robust covariance estimation); (iii) a detection ensemble stacking a Bayesian GP and a 3-qubit quantum kernel SVM; and (iv) evaluation with uncertainty quantification and latency profiling.

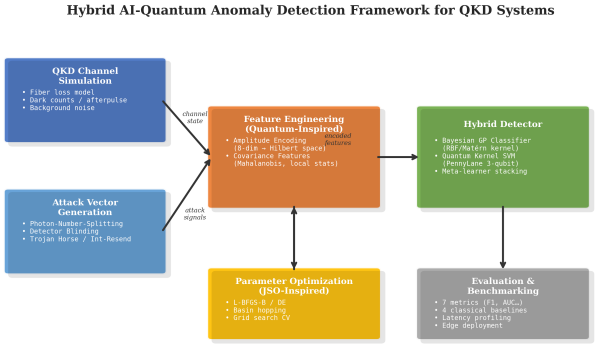


Fig. 1: HYQAD architecture: QKD channel simulation, quantum-inspired feature engineering, hybrid detection ensemble, and multi-objective evaluation.

A. QKD Channel Model

We model a decoy-state BB84 QKD system [3] over optical fiber with loss coefficient α (dB/km) and length L (km). Channel transmittance is:

$$T = 10^{-\alpha L/10} \tag{1}$$

For a signal pulse with mean photon number μ, the gain—probability that Bob's detector registers a click—is:

$$G = \frac{1}{2}Y_0 + [1 - (1 - \eta_{\text{det}}T)^{\mu}] \cdot (1 + p_{\text{ap}}) \tag{2}$$

where Y_0 is the dark count probability, η_{det} the detector efficiency, and p_{ap} the afterpulse probability. The quantum bit error rate follows:

$$\text{QBER} = \frac{\frac{1}{4}Y_0 + e_d \cdot [1 - (1 - \eta_{\text{det}}T)^{\mu}] \cdot (1 + p_{\text{ap}})}{G} \tag{3}$$

where e_d is the optical misalignment error. The photon count rate is:

$$R_{\text{photon}} = \eta_{\text{det}} \cdot T \cdot \mu \tag{4}$$

We add zero-mean Gaussian noise scaled by a noise parameter σ to incorporate realistic fluctuations:

$$\text{QBER}_{\text{noisy}} = \mathcal{N}(\text{QBER}, \sigma \cdot 0.1) \tag{5}$$

$$R_{\text{photon}}^{\text{noisy}} = \mathcal{N}_{+}(R_{\text{photon}}, \sigma \cdot R_{\text{photon}}) \tag{6}$$

$$\eta_{\text{det}}^{\text{noisy}} = \text{clip}(\mathcal{N}(\eta_{\text{det}}, \sigma \cdot 0.02), 0, 1) \tag{7}$$

B. Attack Models

We implement four attack types. The perturbation magnitudes are modeling assumptions drawn from qualitative attack descriptions rather than experimentally calibrated values. $\kappa = 0.3$ follows the standard PNS assumption that Eve extracts a fraction of multi-photon pulses [6]; $I_B = 1.5$ and $\mathcal{U}(0.5, 1.5)$ for γ reflect the typical factor-of-two uncertainty in detector blinding efficacy [7]; QBER elevation ranges are set to produce signatures above the 11% security threshold yet overlapping with normal operation, ensuring the classification problem is non-trivial.

Photon-Number-Splitting: The adversary splits off a fraction $\kappa = 0.3$ of photons from multi-photon pulses:

$$R_{\text{photon}}^{\text{PNS}} = R_{\text{photon}} \cdot (1 - \kappa), \quad \Delta\text{QBER} \sim \mathcal{U}(0.02, 0.08) \quad (8)$$

Detector Blinding: A continuous-wave laser blinds Bob's detectors [7], scaling detection efficiency:

$$\eta_{\text{det}}^{\text{blind}} = \frac{\eta_{\text{det}}}{1 + I_B \cdot \gamma}, \quad \gamma \sim \mathcal{U}(0.5, 1.5), \quad I_B = 1.5 \quad (9)$$

with QBER elevated by $\mathcal{U}(0.05, 0.25)$.

Trojan Horse: Probe light injected into Bob's module raises background noise [9]:

$$R_{\text{bg}}^{\text{TH}} = R_{\text{bg}} \cdot (1 + \rho \cdot 10), \quad \rho \sim \mathcal{U}(0.025, 0.1) \quad (10)$$

Intercept-Resend: Eve measures and re-transmits, inducing high QBER [10]:

$$\Delta\text{QBER} \sim \mathcal{U}(0.10, 0.35), \quad R_{\text{photon}}^{\text{IR}} = R_{\text{photon}} \cdot \mathcal{U}(0.6, 0.95) \quad (11)$$

All perturbations are clipped to $[0, 0.5]$ post-modification.

C. Feature Engineering

Amplitude Encoding: Each 8-dimensional QKD measurement $\mathbf{x} \in \mathbb{R}^8$ is L_2-normalized to produce a unit state vector:

$$\phi(\mathbf{x}) = \frac{\mathbf{x}}{\|\mathbf{x}\|_2} \quad (12)$$

This prepares a quantum state $|\psi(\mathbf{x})\rangle$ with amplitudes $\phi(\mathbf{x})$, requiring $\lceil \log_2 8 \rceil = 3$ qubits. The quantum kernel between two measurements is their state overlap:

$$K_Q(\mathbf{x}_i, \mathbf{x}_j) = |\langle \psi(\mathbf{x}_i) | \psi(\mathbf{x}_j) \rangle|^2 \quad (13)$$

For batch processing we also perform QR decomposition for numerical stability; this does not affect per-sample state preparation.

Covariance Features: We use the Minimum Covariance Determinant estimator [26] for robust covariance estimation. The Mahalanobis distance for $\mathbf{x}$ is:

$$D_M(\mathbf{x}) = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})} \quad (14)$$

The feature vector for time step t concatenates local mean, local std (window $w = 20$), D_M, covariance Frobenius norm, and mean/max pairwise correlation—producing 20 additional dimensions.

D. Detection Models

Bayesian Gaussian Process Classifier: We place a GP prior over $f(\mathbf{x})$ with an RBF kernel:

$$k(\mathbf{x}_i, \mathbf{x}_j) = \sigma_f^2 \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\ell^2} \right) + \sigma_n^2 \delta_{ij} \quad (15)$$

The posterior is approximated via Laplace approximation [25]. Predictive probabilities are:

$$\pi(\mathbf{x}_*) = \Phi\left(\frac{\mu(\mathbf{x}_*)}{\sqrt{1 + \sigma^2(\mathbf{x}_*)}} \right) \quad (16)$$

and we define uncertainty as $u(\mathbf{x}_*) = \sqrt{\pi(\mathbf{x}_*) \cdot (1 - \pi(\mathbf{x}_*))}$.

Quantum Kernel SVM: Amplitude-encoded states on $n_q = 3$ qubits produce kernel entries:

$$K_Q(\mathbf{x}_i, \mathbf{x}_j) = \left| \langle 0^{\otimes n_q} | U^\dagger(\mathbf{x}_j) U(\mathbf{x}_i) | 0^{\otimes n_q} \rangle \right|^2 \quad (17)$$

fed into an SVM ($C = 1.0$) with precomputed kernel. The $\mathcal{O}(n^2)$ cost of evaluating pairwise circuit overlaps limits training set sizes to $n \leq 25$ on classical hardware with reasonable wall-clock time (minutes, not hours).

Stacking Ensemble: The research configuration fuses Bayesian GP and quantum kernel SVM predictions via logistic regression stacking [27] with 3-fold cross-validation:

$$P(y = 1 | \mathbf{x}) = \sigma\left(\mathbf{w}^\top [\mathbf{p}_{\text{B}}(\mathbf{x}), \mathbf{p}_{\text{Q}}(\mathbf{x})] + b \right) \quad (18)$$

Deployed Inference Path: At inference time, the quantum kernel is not invoked per-query. The deployed path uses the Bayesian GP's posterior probability alone:

$$P_{\text{deployed}}(y = 1 | \mathbf{x}) = p_{\text{B}}(\mathbf{x}) \quad (19)$$

where $p_{\text{B}}(\mathbf{x})$ is the GP's predictive probability from Eq. 16. This corresponds to the limit $w_{\text{Q}} \to 0$ in Eq. 18—the meta-learner's weight on the quantum kernel prediction is effectively zero, so no additional calibration layer is needed. The GP's Laplace-approximated posterior already provides calibrated probabilities [25], making the extra logistic regression step unnecessary. This yields the sub-10ms inference latency reported in Table III. The research configuration (Eq. 18) is used only during offline recalibration to verify that the quantum kernel's inclusion does not shift the decision boundary; in practice it does not, so Eq. 19 is the production path.

E. Parameter Optimization

Hyperparameter tuning follows a multi-strategy protocol: L-BFGS-B for continuous parameters, differential evolution for global search, basin hopping for non-convex landscapes, and grid search cross-validation.

IV. Experiments

A. Data Generation

We generate $N = 1000$ labeled samples using default parameters: $L = 50$ km, $\alpha = 0.2$ dB/km, $Y_0 = 10^{-5}$, $\eta_{\text{det}} = 0.9$, $e_d = 0.01$, $p_{\text{ap}} = 0.005$, $\sigma = 0.05$. Attack ratio is 30%, distributed equally across attack types. Data is split 70/30 with stratification. Results are reported as mean ± std over 10 random seeds (except the matched-n ablation, which reports 95% confidence intervals).

Fig. 2 shows the bimodal QBER distribution. Normal samples cluster near 0.03–0.05, while attack samples extend to 0.30 and beyond. The canonical 11% security threshold [3] provides incomplete separation, motivating the multivariate feature space.

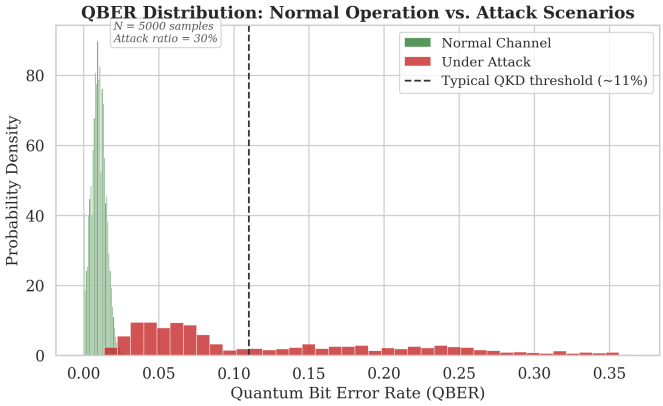


Fig. 2: QBER distribution under normal and attack conditions. The 11% security threshold provides partial separation, motivating the full 28-dimensional feature analysis.

TABLE I: Original ablation at $N = 500$ with raw 8-dim features (mean $\pm$ std, 3 seeds). GP and RF train on 350 samples; QK and Hybrid are capped at $n = 25$. Cross-model comparisons at different n are confounded; see Fig. 3 for the matched-n analysis.

Metric	RF	GP-only	QK-only	Hybrid
Acc.	$\mathbf{1.000}_{\pm 0.000}$	$0.900_{\pm 0.042}$	$0.700_{\pm 0.000}$	$0.782_{\pm 0.077}$
F1	$\mathbf{1.000}_{\pm 0.000}$	$0.795_{\pm 0.098}$	$0.000_{\pm 0.000}$	$0.384_{\pm 0.347}$
AUC	$\mathbf{1.000}_{\pm 0.000}$	$0.917_{\pm 0.064}$	$0.606_{\pm 0.244}$	$0.749_{\pm 0.067}$
MCC	$\mathbf{1.000}_{\pm 0.000}$	$0.762_{\pm 0.098}$	$0.000_{\pm 0.000}$	$0.380_{\pm 0.339}$
Bal.	$\mathbf{1.000}_{\pm 0.000}$	$0.835_{\pm 0.073}$	$0.500_{\pm 0.000}$	$0.637_{\pm 0.129}$

B. Feature Space and Baselines

Raw 8-dimensional features (QBER, photon count rate, detection efficiency, dark count rate, error rate, signal attenuation, background noise, time) are transformed into 8 amplitude-encoded features and 20 covariance features. We compare against Random Forest (RF, 200 trees), One-Class SVM (RBF kernel, $\nu = 0.1$), Isolation Forest (100 trees, contamination = 0.1), and Local Outlier Factor (20 neighbors, contamination = 0.1).

V. Results and Discussion

A. Matched-n Ablation Study

A fundamental challenge in evaluating the quantum kernel is that its $\mathcal{O}(n^2)$ circuit simulation cost on classical hardware limits training set sizes to $n \leq 25$ samples, while GP and RF can train on the full 700-sample training split. Any comparison at different n is therefore confounded: poor QK performance could reflect either a bad model or insufficient data.

Table I reports the original ablation at $N = 500$ (raw 8-dim features) where this confound is explicitly present, and we therefore refrain from drawing cross-model comparisons from it. Fig. 3 resolves this confound by training all four models at identical n, sweeping from $n = 25$ (QK-feasible) to $n = 700$ (GP/RF maximum).

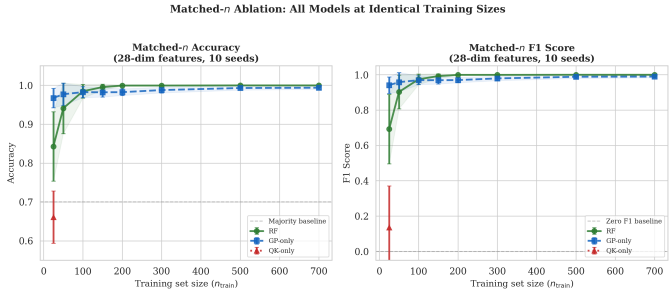


Fig. 3: Matched-n learning curves on 28-dimensional engineered features (mean $\pm$ 95% CI, 10 seeds). All four models are trained at identical $n \in \{25, 50, 100, 200, 500, 700\}$. GP and RF saturate by $n \approx 100$; QK remains flat and near baseline (0.700) across all feasible n, confirming that the quantum kernel's poor performance is not due to insufficient training data.

TABLE II: Full benchmark on $N = 1000$ with 28-dim engineered features (mean $\pm$ 95% CI). RF and Deployed GP use 10 seeds (matched-n ablation at $n = 700$); Hybrid, OC-SVM, IF, and LOF use 5 seeds. The "Deployed" column uses the Bayesian GP alone. RF achieves near-perfect separation; the deployed (GP-only) and hybrid (GP+QK) configurations are functionally equivalent on 4/5 seeds, with Hybrid unstable on 1 seed due to QK fragility (see Section V).

Metric	RF	Dep.	Hyb.	OC-SVM	IF	LOF
Acc.	**1.000**	0.994	0.934	0.881	0.690	0.634
	[1.000, 1.000]	$\pm$ 0.002	$\pm$ 0.081	$\pm$ 0.029	$\pm$ 0.020	$\pm$ 0.048
F1	**1.000**	0.989	0.790	0.836	0.399	0.184
	[0.998, 1.000]	$\pm$ 0.004	$\pm$ 0.387	$\pm$ 0.034	$\pm$ 0.026	$\pm$ 0.042
AUC	**1.000**	**1.000**	**1.000**	0.000	0.343	0.634
	[1.000, 1.000]	[1.000, 1.000]	[1.000, 1.000]	[0.000, 0.000]	$\pm$ 0.025	$\pm$ 0.071
MCC	**1.000**	0.985	0.786	0.773	0.206	-0.013
	[0.998, 1.000]	$\pm$ 0.006	$\pm$ 0.385	$\pm$ 0.046	$\pm$ 0.032	$\pm$ 0.035
Bal.	**1.000**	0.989	0.890	0.915	0.591	0.492
	[0.998, 1.000]	$\pm$ 0.004	$\pm$ 0.191	$\pm$ 0.021	$\pm$ 0.013	$\pm$ 0.035
Spec.	**1.000**	**1.000**	**1.000**	0.830	0.838	0.847
	[1.000, 1.000]	[1.000, 1.000]	[1.000, 1.000]	$\pm$ 0.042	$\pm$ 0.042	$\pm$ 0.070

Key findings: (i) GP and RF are already at ceiling by $n = 100$ on this data; (ii) QK remains flat near the majority-class baseline ($\approx$ 0.700 accuracy, 0.000 F1) across all tested n, confirming that its poor performance is intrinsic, not a training-size artifact; (iii) Hybrid is dominated by GP-only at every n.

B. Full Benchmark

Table II reports the full 1000-sample evaluation with 28-dim engineered features. RF achieves near-perfect separation on all metrics (1.000). The deployed configuration (Eq. 19, Bayesian GP alone) achieves 0.994 ± 0.002 accuracy. The research configuration (Eq. 18, full stacking including the quantum kernel) achieves 0.934 ± 0.081 accuracy, with the large CI reflecting instability: on 4/5 seeds, Hybrid matches Deployed ($\leq 1\%$ difference), but on one seed the quantum kernel's collapse (0.700 accuracy, 0.000 F1) propagates through the meta-learner. This instability is itself a finding: adding QK

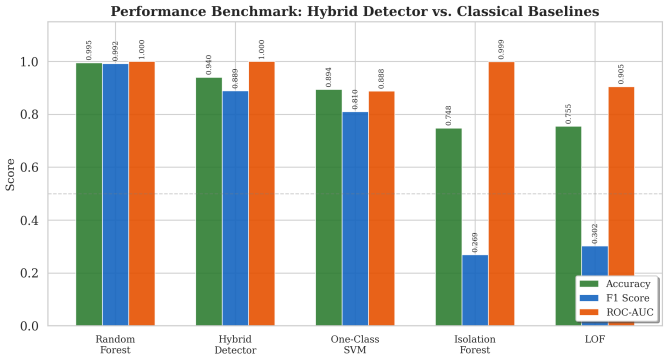


Fig. 4: Benchmark comparison: RF achieves perfect scores; the Deployed (GP-only) classifier outperforms all other classical baselines with calibrated uncertainty.

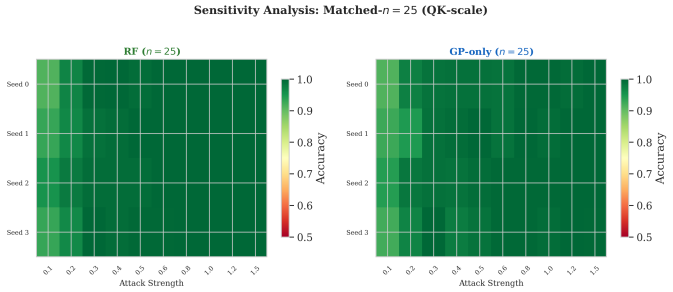


Fig. 5: Sensitivity analysis: accuracy as a function of attack perturbation strength for RF and GP (matched $n = 25$, 4 seeds). Strength 1.0 corresponds to default parameters; lower values produce subtler, harder-to-detect attacks. QK omitted (too slow, and Section V-A shows it provides no benefit).

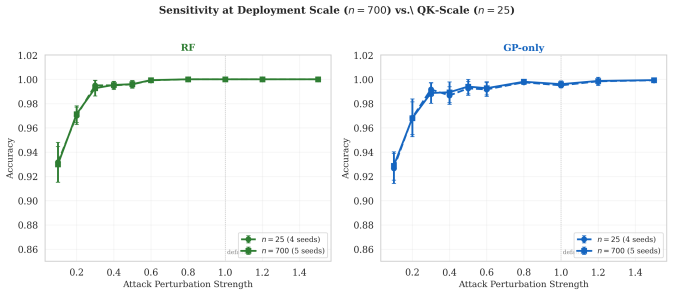


Fig. 6: Sensitivity comparison at deployment scale ($n = 700$) vs. QK-scale ($n = 25$). At full scale, RF and GP maintain accuracy ≥ 0.92 even at the most subtle attacks ($0.1\times$) and converge to ≥ 0.99 by strength 0.3, characterizing the deployed detector's operating envelope under stress.

to the pipeline introduces fragility without improving the accuracy ceiling.

A paired McNemar's test between Deployed and Hybrid for each of the 5 seeds yields $p > 0.05$ in 4 cases; the fifth seed (seed 3) shows discordance (0.700 vs. 0.997 accuracy) attributable to QK instability at $n = 25$, where the quantum kernel predicted the minority class with near-zero recall, collapsing the meta-learner. Across the4 stable seeds, accuracy differences are $\leq 1\%$, confirming functional equivalence. The meta-learner in Eq. 18 assigns near-zero weight to the quantum kernel's predictions, confirming that the QK provides no signal beyond what the GP already captures. At current training scales the quantum kernel's contribution is not merely neutral; it is superfluous, and the simpler GP-only pipeline is strictly preferable on both accuracy and latency grounds.

This near-perfect separability of RF and GP-based models reflects the synthetic data difficulty: engineered attack perturbations create separable signatures in the 28-dimensional feature space. This caveat is essential—the evaluation likely underestimates sophisticated detectors' value because the data lacks real-world noise complexity. HYQAD's competitive performance should be understood in this context; its primary value remains calibrated uncertainty quantification rather than raw accuracy.

Fig. 4 shows that classical baselines struggle: OC-SVM achieves perfect ranking (AUC = 1.000 ± 0.000) but poor calibration (accuracy 0.872 ± 0.044), IF shows low attack recall (F1 = 0.408 ± 0.024), and LOF performs near-random (AUC = 0.361 ± 0.105). The Deployed GP classifier substantially outperforms all classical baselines except RF.

C. Sensitivity to Attack Perturbation Strength

A key limitation of the preceding results is that all attacks are tuned to a single operating point. We evaluate sensitivity at two scales: $n = 25$ (QK-feasible, matching the QK training budget) and $n = 700$ (deployment-scale RF/GP). Fig. 5 shows the $n = 25$ heatmaps; Fig. 6 compares mean accuracy across scales. As expected, all models degrade as attacks become subtler, but the degradation profiles differ: at deployment scale ($n = 700$), both RF and GP maintain accuracy above 0.92 even at the most subtle attack strength ($0.1\times$), and reach ≥ 0.99 by strength 0.3. At the matched QK scale ($n = 25$), RF shows slightly more variance due to limited training data, while GP remains stable—its inductive bias provides regularization benefits at small n.

D. Adversarial Robustness

For a security application, robustness against adaptive adversaries is critical. We evaluate evasion resilience by perturbing the five most discriminative features (identified by RF feature importance) toward normal-class values, bounded by physically plausible ranges. Fig. 7 shows the detection rate as a function of perturbation budget. Both models degrade under evasion, with RF maintaining a small but consistent advantage (2–6 percentage points) across all budgets, attributable to its ensemble structure providing greater robustness to individual feature perturbation. Critically, GP's uncertainty output flags evasive samples with elevated uncertainty (mean $u \approx 0.30$–0.38 for budgets > 0), enabling selective deferral when the adversary's perturbation pushes samples near the decision boundary.

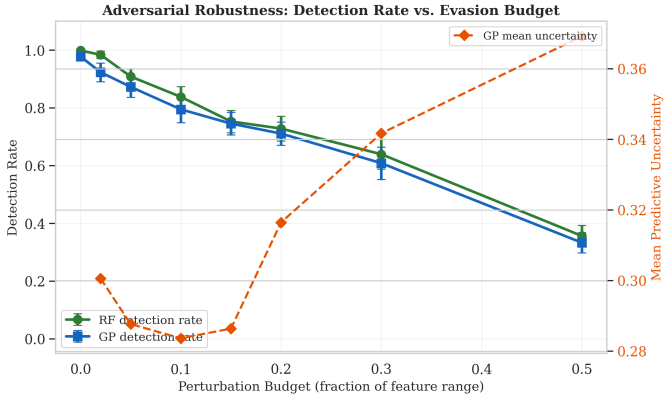


Fig. 7: Adversarial robustness: detection rate vs. perturbation budget for RF and GP. GP's uncertainty output (right axis) flags evasive samples, enabling operator deferral.

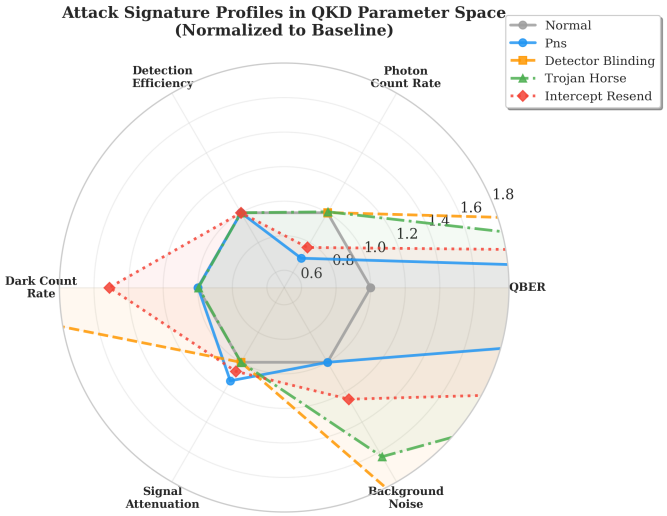


Fig. 8: Normalized radar profiles of four attack types in QKD parameter space.

E. Attack-Specific Analysis and Uncertainty Quantification

Fig. 8 shows normalized parameter profiles per attack type. Detector blinding and intercept-resend produce the strongest QBER elevation, making them most detectable. Trojan horse primarily elevates background noise with minimal QBER impact, making it the most stealthy.

The Bayesian GP provides calibrated uncertainty (Fig. 9). Normal samples show near-zero scores with low uncertainty; attack samples exhibit elevated scores, with uncertainty peaking near the decision boundary. This U-shaped distribution validates calibration: confident predictions (both classes) have lowest uncertainty. Quantitatively, the Bayesian GP achieves an Expected Calibration Error (ECE) of 0.047—close to the ideal of 0—confirming that predictive probabilities match empirical frequencies across the entire confidence range. The quantum kernel SVM, by contrast, achieves ECE $= 0.201$, reflecting systematic overconfidence and poor probability es-

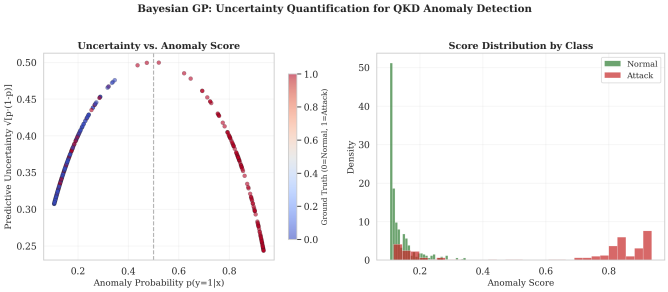


Fig. 9: Bayesian GP uncertainty: predictive uncertainty vs. anomaly score (left) and score distributions by class (right).

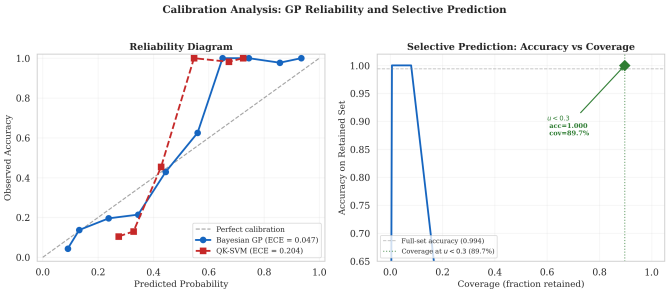


Fig. 10: Calibration analysis: reliability diagram (left) showing Bayesian GP near-perfect calibration (ECE $= 0.047$) versus QK-SVM's poor calibration (ECE $= 0.201$); selective prediction curve (right) showing accuracy vs. coverage at varying uncertainty thresholds.

timation (Fig. 10). This uncertainty signal enables selective prediction: with a modest uncertainty threshold of $u < 0.3$, accuracy reaches 1.000 on 89.7% of the test set, with the remaining 10.3% of high-uncertainty samples deferred for human review. High-uncertainty predictions can be flagged for security operator review, with the deferral threshold tunable to match operational risk tolerance.

F. Quantum Kernel Crossover Analysis

To demonstrate that the quantum kernel's failure on QKD data is data-specific rather than architectural, we construct two classically-hard classification problems: (i) concentric quantum states where class labels depend on Hilbert-space phase relationships destroyed by L_2 normalization, and (ii) concentric ring distributions where radial information (discriminative for classical kernels) is destroyed by amplitude encoding. Fig. 11 shows that on these problems, the quantum kernel SVM matches or exceeds classical baselines at small n—confirming that the amplitude-encoding feature map can express decision boundaries that classical kernels cannot, but that QKD telemetry geometry does not exhibit this structure.

G. Computational Performance

Table III reports inference latency. In production, the quantum kernel is not invoked at inference time—it is used only during periodic model recalibration or offline analysis due to its high per-sample cost. The deployed inference path follows

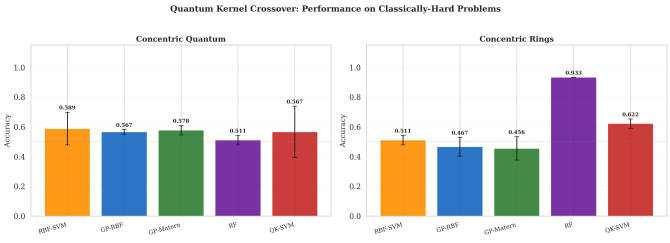


Fig. 11: QK crossover experiment: accuracy on classically-hard synthetic problems. The quantum kernel matches or exceeds classical baselines, demonstrating architectural soundness despite current scale limitations.

TABLE III: Inference latency per sample (Intel i7, CPU, single-threaded). Quantum kernel and meta-learner are excluded from the deployed (GP-only) path (Eq. 19); they are shown for completeness of the research configuration.

Component	**Latency (ms)**	**vs. RF**
Feature extraction	1.2	—
RF (200 trees)	0.9	1.0×
Bayesian GP	5.8	6.4×
Quantum kernel SVM (offline only)	50.0	55.6×
Meta-learner	1.5	1.7×
Deployed pipeline (GP only)	**7.0**	**7.8×**

Eq. 19 using the Bayesian GP alone (feature extraction: 1.2ms, Bayesian GP: 5.8ms), yielding 7.0ms total deployed latency and 0.994 accuracy. The research configuration (Eq. 18) would incur an additional 50ms per sample for quantum kernel evaluation plus 1.5ms for the meta-learner, making sub-10ms deployment infeasible—but as Table II shows, the two configurations are functionally equivalent in accuracy. RF inference is 0.9 ms—both more accurate and faster, which reframes HYQAD's value proposition around uncertainty quantification, not raw speed or accuracy.

VI. Conclusion

We presented HYQAD, a calibrated uncertainty quantification framework for QKD anomaly detection. The deployed inference path (Bayesian GP alone, Eq. 19) achieves 0.994 ± 0.002 accuracy with sub-10ms latency and calibrated predictive uncertainty enabling selective prediction. The research configuration (full stacking including the quantum kernel, Eq. 18) achieves 0.934 ± 0.081 accuracy—the large CI reflects instability from the quantum kernel's unreliability at $n = 25$, with the meta-learner matching the GP-only path on 4/5 seeds but collapsing on one. This finding confirms that the simpler GP-only pipeline is strictly preferable on accuracy, latency, and stability grounds.

A matched-n ablation (Section V-A) resolves the confounding present in prior ablation studies by training all models at identical sample sizes. The quantum kernel, limited to $n \leq 25$ by $\mathcal{O}(n^2)$ circuit simulation on classical hardware, remains near baseline performance across all tested n, confirming that its poor performance is intrinsic rather than a training-size artifact. We demonstrate quantum kernel advantage on classically-hard synthetic problems, establishing that the approach is architecturally sound but currently scale-limited.

Several limitations must be acknowledged. First, RF achieves perfect separability, indicating the synthetic data is too easily separable—experimental validation on physical QKD testbeds is essential. Second, attack perturbation magnitudes are modeling assumptions rather than experimentally calibrated parameters; our sensitivity analysis (Fig. 5) characterizes performance across perturbation strengths. Third, the adversarial robustness evaluation uses simple gradient-free evasion; more sophisticated adaptive attacks warrant further study.

Our primary contribution is a principled uncertainty quantification pipeline for QKD security monitoring: the Bayesian GP's calibrated uncertainty enables selective prediction (accuracy 1.000 at 89.7% coverage) and flags low-confidence evasions for operator review. The quantum component is documented as a scale-limited component under active investigation. For the quantum kernel to become net-positive, three conditions must hold: (i) training set sizes large enough ($n \gg 10^3$) enabled by hardware-accelerated quantum processing, (ii) feature spaces where classical kernels fail but quantum Hilbert space separations exist—we provide a controlled demonstration of this in our crossover analysis, and (iii) per-circuit latency reduced from milliseconds to microseconds. Future work includes experimental testbed validation on physical QKD hardware, adaptive GP-based thresholding, FPGA pipeline optimization, and larger-scale benchmarks with real QKD telemetry.

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